SURAJ PRASAD KALAUNI

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EDUCATION

Delhi Technological University — Delhi, India
M.Tech Software Engineering, GPA: 8.33 — *Aug 2024 - Aug 2026*
Dr. A.P.J Abdul Kalam Technological University — Lucknow, India
B.Tech Computer Science and Engineering, GPA: 8.12 — *Jun 2020 - Mar 2024*

SKILLS SUMMARY

- **Languages**: Python, C++, C, Java, Golang
- **ML Frameworks & Inference Tools**: PyTorch, Hugging Face Transformers, TensorFlow, vLLM, BitsAndBytes, ONNX, LangChain, SGLang
- **Inference & Optimization**: Speculative Decoding, INT8/FP16 Quantization, Structured Pruning & Compression, Sparse Attention, KV-Cache Optimization, Post-Training Techniques, RAG, Prompt Engineering
- **Systems & Infrastructure**: Docker, Kubernetes, AWS (EC2, S3, Lambda), Linux, REST APIs, CI/CD, Observability & Monitoring
- **Core Computer Science**: Data Structures & Algorithms, Operating Systems, Computer Networks, System Design, OOP, DBMS

WORK EXPERIENCE

Samsung Research and Development | Software Engineer Intern — July 2025 – January 2026

- Engineered LLM-powered agentic workflows integrated into internal developer tools; applied prompt chaining and tool-use patterns to automate multi-step developer tasks, cutting manual overhead significantly.
- Optimised transformer-based inference pipelines for production deployment — applied batching, prompt compression, and caching strategies, reducing average inference latency by **15%** on internal benchmarks.
- Built and containerised full-stack services (Next.js, TypeScript) using Docker & Kubernetes on AWS, achieving **99.9%** uptime with improved system observability and monitoring.
- Collaborated with cross-functional ML and platform teams to validate model outputs, debug inference failures, and tune generation parameters across multi-agent orchestration frameworks.

PROJECTS

Speculative Decoding — LLM Inference Acceleration — 2025

- Implemented speculative decoding from scratch using GPT-2 as a draft model and GPT-2 XL as the target verifier; reproduced the token acceptance and rejection mechanism as described in Leviathan et al. (2023).
- Measured **2.1× end-to-end throughput speedup** over standard autoregressive decoding at equivalent output quality; analysed the acceptance rate vs. speedup tradeoff across temperature settings.
- Profiled token generation latency, KV-cache memory usage, and batch throughput across sequence lengths to quantify performance gains on CPU and single-GPU environments.

LLM Inference Benchmarking & Quantization Pipeline — 2025

- Built an end-to-end benchmarking tool for open-source LLMs (LLaMA 3.2-1B, Mistral-7B) measuring tokens/second, per-token latency, peak memory, and batch throughput under varying load.
- Applied INT8 post-training quantization via BitsAndBytes; achieved **38% reduction in GPU memory** with less than 1.2% perplexity degradation on WikiText-2 compared to the FP16 baseline.
- Exported quantized models to ONNX format and validated inference outputs, documenting latency-accuracy tradeoffs across precision levels (FP32, FP16, INT8) for reproducible evaluation.

Transformer Pruning & Compression for Efficient Inference — 2024

- Designed a structured pruning pipeline for DistilBERT; applied magnitude-based attention-head pruning at 10%, 20%, and 30% sparsity levels and measured downstream accuracy on SST-2 classification.
- Achieved **27% reduction in inference latency** at 20% sparsity with only 0.8% accuracy drop; benchmarked against a knowledge distillation baseline to compare compression strategies.
- Analysed sparsity-accuracy tradeoff curves and visualised attention head importance scores to identify redundant heads — findings directly informed pruning threshold selection.

PUBLICATIONS

- Retinal Fundus Image and Deep Learning: A Novel Approach for Cardiovascular Risk Prediction, Published in AIP Conference Proceedings, 2025.
- Introduced a novel method for cardiovascular risk prediction, utilizing retinal fundus images and deep learning techniques, achieving 79.41% accuracy in classifying cardiovascular risk.
- DOI Link: https://doi.org/10.1063/5.0247009.

ACHIEVEMENTS

- GATE CSE 2024: Qualified GATE CSE 2024 with Marks: 40.48, Score: 490/1000, Rank: 5943 among 123967.
- GATE DA 2024: Qualified GATE DA 2024 with Marks: 38, Score: 362/1000, Rank: 5973 among 39210.